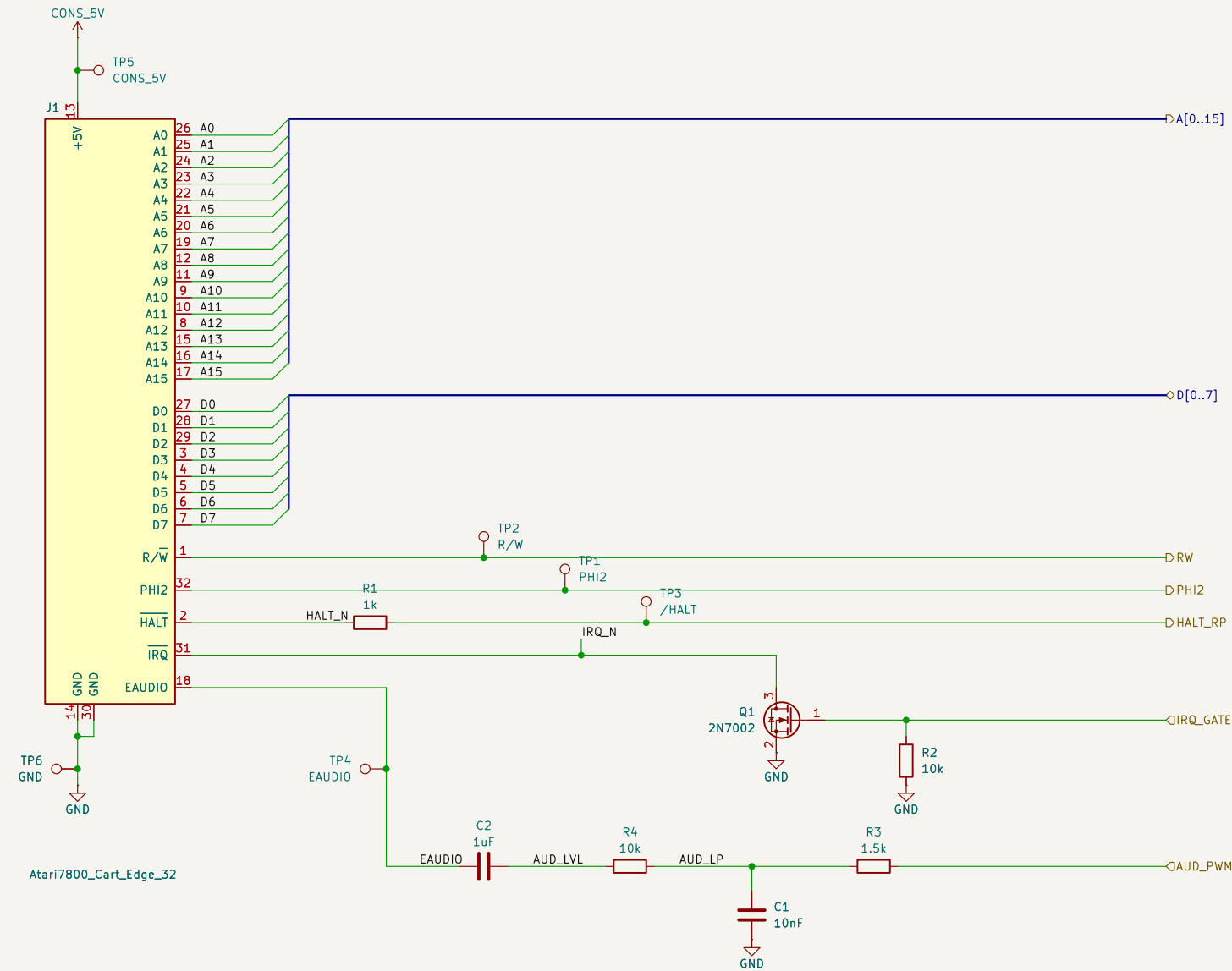
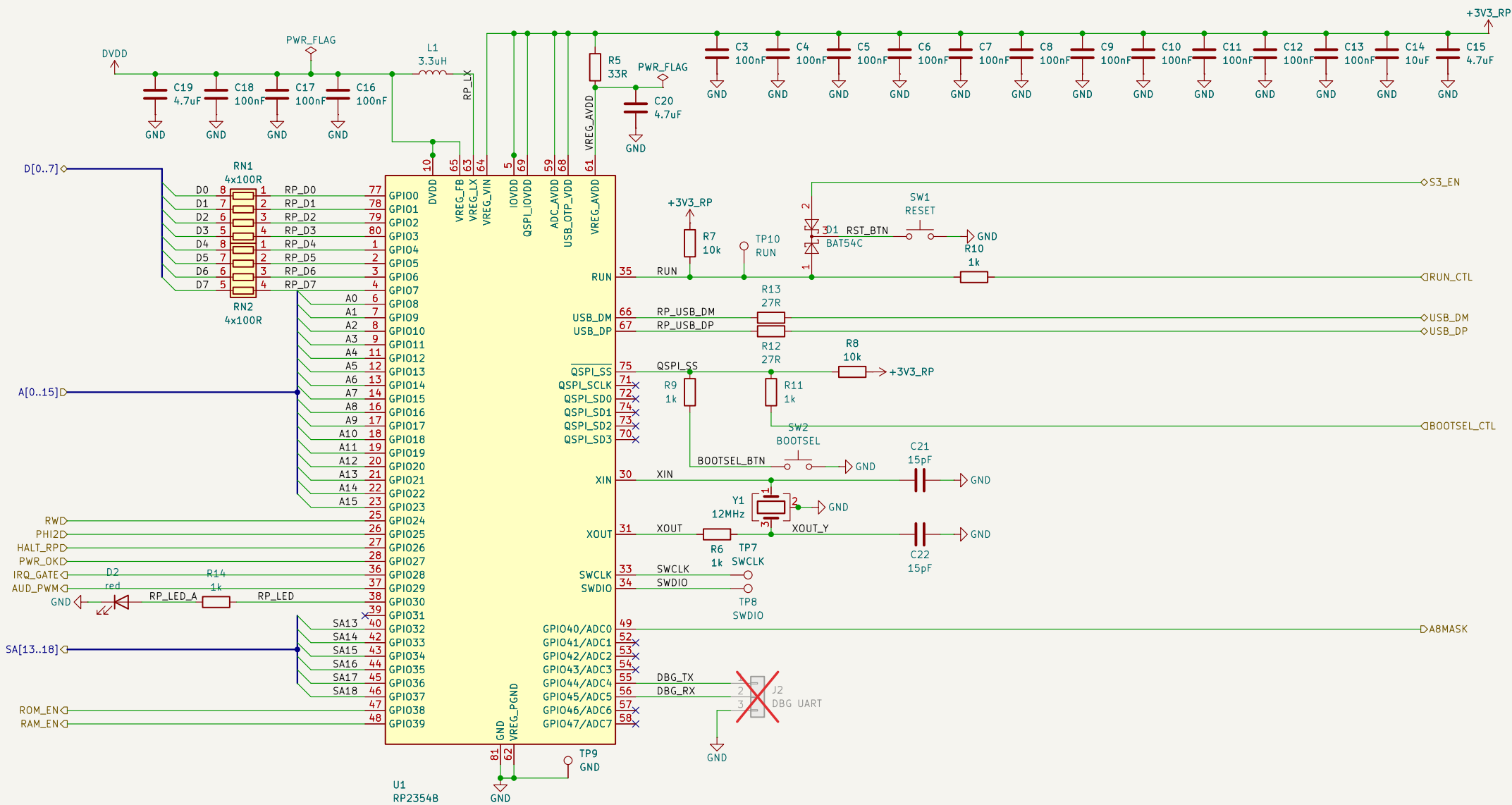


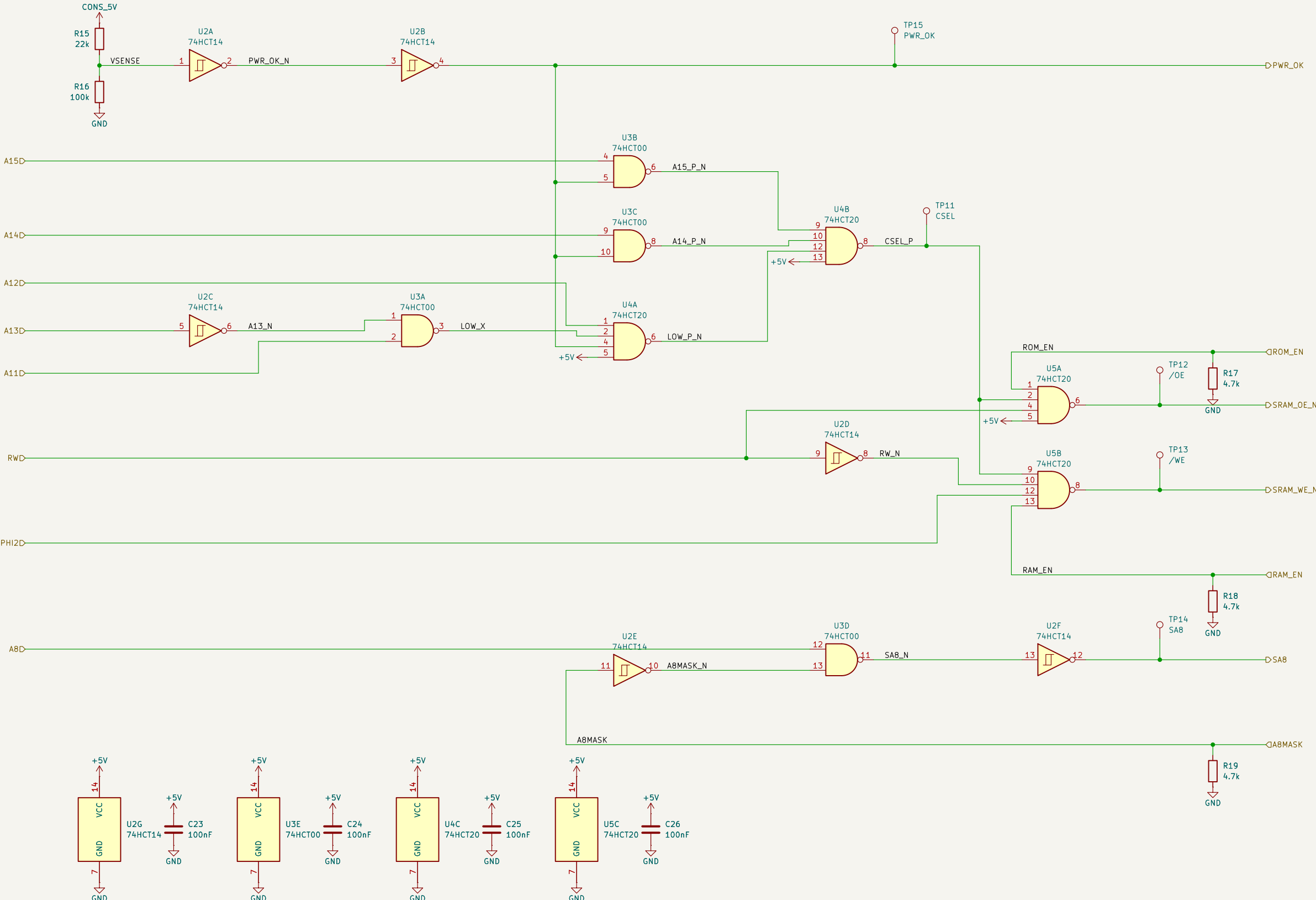
Atari 7800 32-pin cartridge slot, drawn by function (finger numbers on the pins). 18 positions at 2.54 mm per face; positions 3 and 16 are key slots in the board for the console connector's plastic keys. Pins 1–16 on F.Cu, the component side, which faces the console's REAR; pin 1 at the left seen from F.Cu, fingers down; pin k over pin 33–k on B.Cu (tools/edge-geom.py, tools/audit/edge_orientation.py). No chip select and no reset on this edge: the glue decodes A15–A11. Cart audio: GPIO29 PWM (833 kHz carrier) → 1.5k / 10n (fc 10.6 kHz) → 10k level → 1u → EAUDIO; the console adds 1.0u + 6.8k into its audio sum (C10, R5), as a POKEY cart's pin 1 pull-up + 12k does. /IRQ: the 2N7002 pulls it low only while GPIO28 is high (the firmware never does); console pull-up R32 2.2k. /HALT (MARIA's weak MOS output) through 1k to the RP: optional. While the console's BIOS is mapped its U4 (74ALS08) decodes A15, A14, A12 low at this edge.



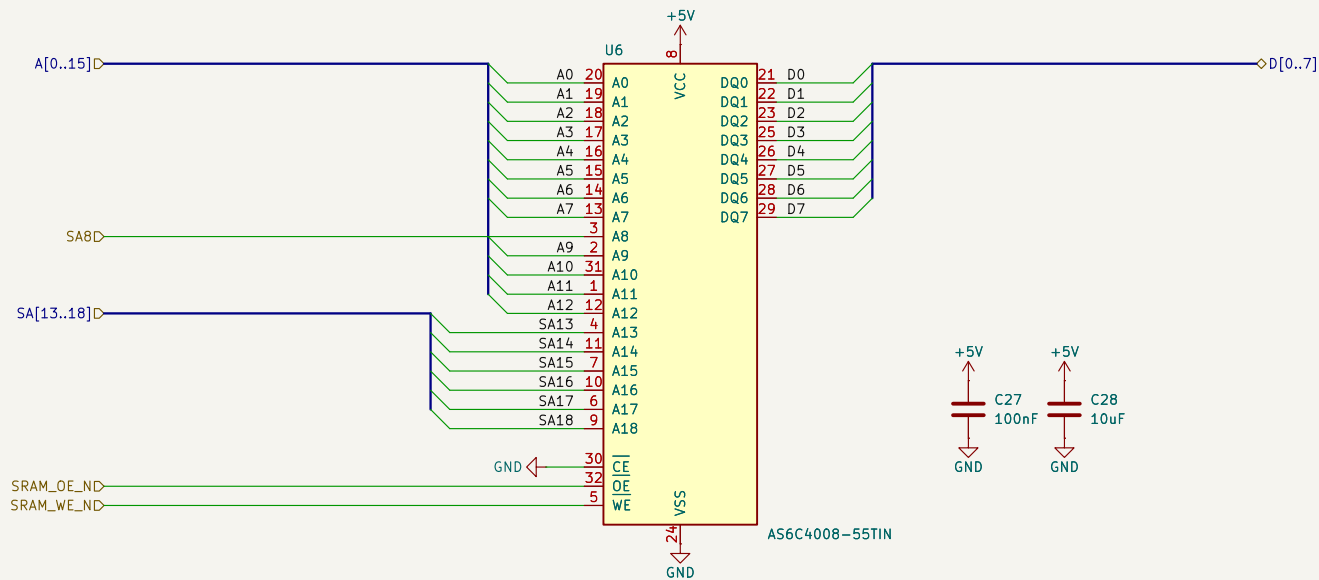
Rev0 audit 2026-10-07: kicad-happy + datasheets, see docs/design-review-rev0.md CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources) Generated from tools/design.py + tools/sch_layout.py - edit those, not this file FujiNet		
Sheet: /edge/ File: edge.kicad_sch		
Title: FujiNet 7800 Rev0 – Cartridge slot, /IRQ, cart audio		
Size: A3	Date: 2026-10-07	Rev: 0
KiCad E.D.A. 10.0.6		Id: 2/8



fujinet-firmware pico/atari-7800 a78_cart.h: D0-D7 GPIO0-7 (through 100R), A0-A15 GPIO8-23, R/W 24, PHI2 25, /HALT 26 (observed only), PWR_OK 27, /IRQ gate 28, audio PWM 29, LED 30: a78map.h slot word on GPIO32-40 (PIO, indexed by A13-A15): SRAM A13-A18, ROM_EN, RAM_EN, A8MASK. GPIO40-47 are not 5 V tolerant: A8MASK drives a 74HCT14 input only, GPIO44/45 the 3.3 V debug header (DNP). One 100 nF per IOVDD pin (key C_IOV<pin>: placed at that pin). RP2350 datasheet 6.3.7: VREG_VIN and IOVDD on one 3.3 V supply (+3V3_RP, the LDO that rises with the console 5V). VREG_AVDD through 33R / 4.7 uF, 3.3 uH from VREG_LX to DVDD (= VREG_FB).



fujinet-firmware pico/atari-7800 a78_cart.h: a78_cartsel(), a78_glue_oe(), a78_glue_we(), a78_glue_a8().
RW = 1 for a read; ROM_EN, RAM_EN, A8MASK = RP GPIO38, 39, 40 (the PIO slot table; 4.7k pull-downs hold them low until it runs, RP2350-E9); PWR_OK = console +5V present (0.82 x CONS_5V into the 74HCT14).
CARTSEL = A15 | A14 | A12 & (A13 | !A11) (\$4000-\$FFFF, \$1000-\$17FF, \$3000-\$3FFF)
CSEL_P = PWR_OK & CARTSEL = NAND(!A15 & PWR_OK), !A14 & PWR_OK, !A12 & LOW_X & PWR_OK))
SRAM /OE = !(CSEL_P & RW & ROM_EN) (no PHI2; MARIA DMA is not PHI2-aligned)
SRAM /WE = !(CSEL_P & !RW & PHI2 & RAM_EN) (PHI2 one gate from /WE); SRAM A8 = A8 & !A8MASK
tools/check_glue.py evaluates this netlist against the firmware's C for every input combination.



A0–A7, A9–A12 and D0–D7 are the console's; SRAM A8 = A8 & !A8MASK (glue sheet); A13–A18 are the RP's slot lines (GPIO32–37); the 8K page for the slot A13–A15 selects. /CE is tied active: /OE and /WE alone decide when the SRAM answers.

Rev0 audit 2026–10–07: kicad-happy + datasheets, see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py – edit those, not this file

FujiNet

Sheet: /sram/

File: sram.kicad_sch

Title: FujiNet 7800 Rev0 – 512K SRAM: AS6C4008

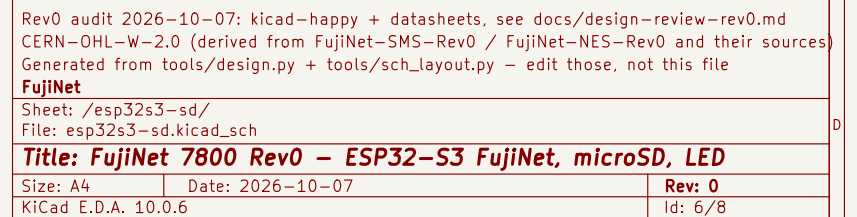
Size: A4

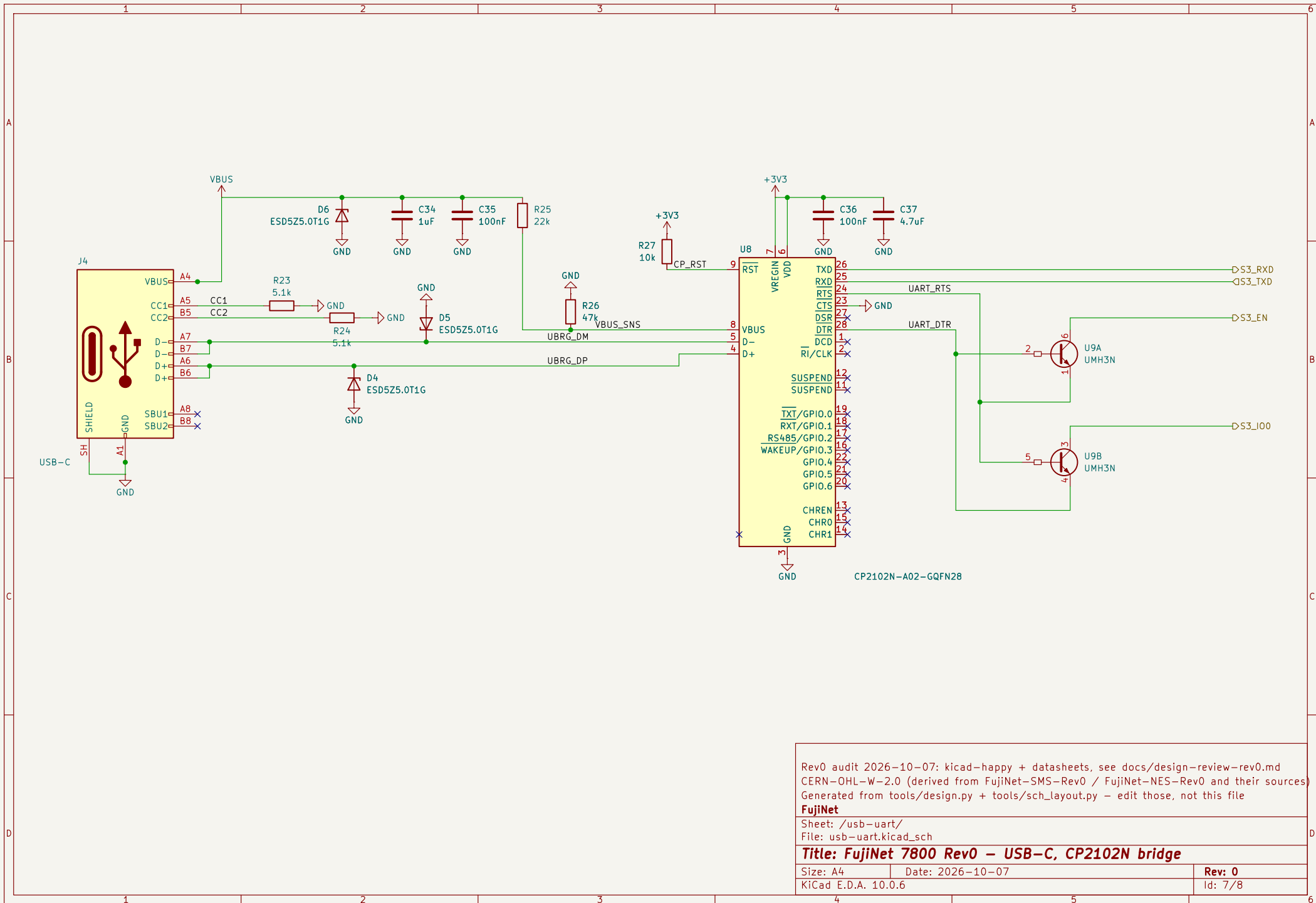
Date: 2026–10–07

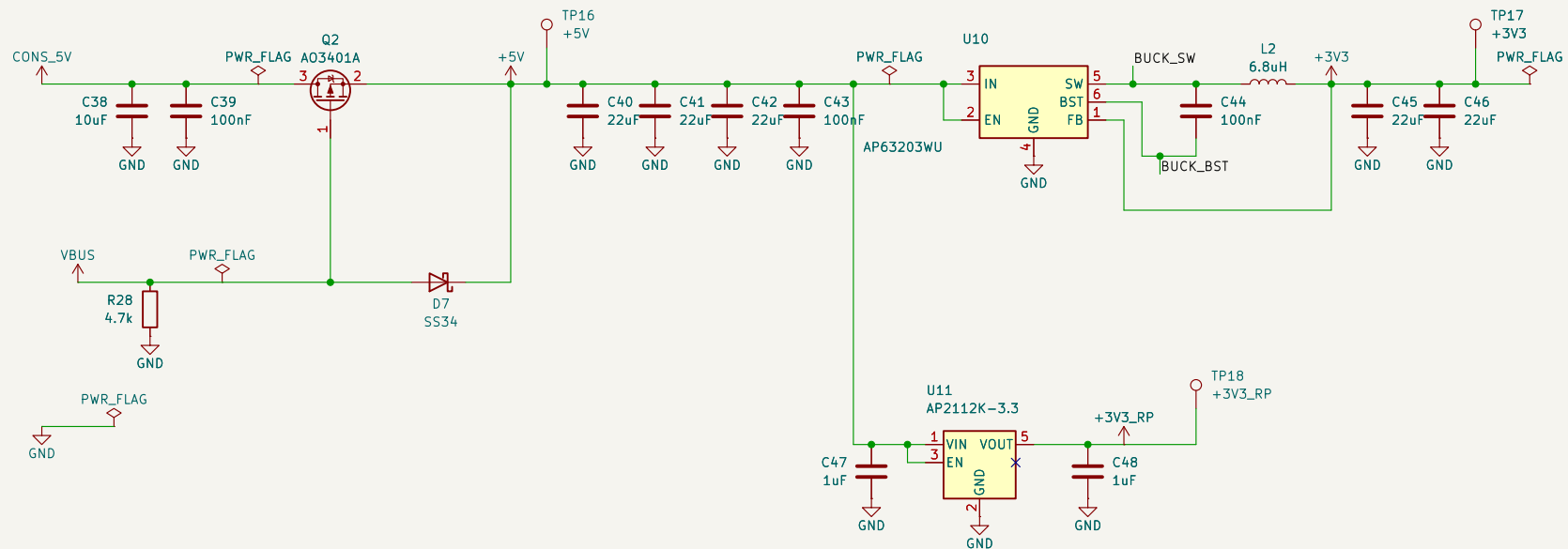
Rev: 0

KiCad E.D.A. 10.0.6

Id: 5/8







Console 5V (slot pin 13, CONS_5V) through the A03401A (gate = VBUS: fully on without USB, body diode only with USB, so nothing back-feeds the console); USB VBUS through the SS34; R_VBPD 4.7k holds the gate low against the SS34's reverse leakage. +5V: SRAM, glue, WS2812, buck, RP LDO. +3V3 (buck): S3, microSD, CP2102N. +3V3_RP (LDO, rises with the 5V rail): every RP2354B supply pin, so its 5V-tolerant pads are powered before the bus is.

Rev0 audit 2026-10-07: kicad-happy + datasheets, see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-SMS-Rev0 / FujiNet-NES-Rev0 and their sources)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet 7800 Rev0 - Power: 5V OR, 3.3V buck, RP LDO

Size: A4 Date: 2026-10-07

KiCad E.D.A. 10.0.6

Rev: 0

Id: 8/8